

Question Paper Code : AD349I — Predicted 2026

B.E./B.Tech. Degree Examinations, November/December 2026

Third/Fourth Semester · Artificial Intelligence and Data Science

AD 349I — Fundamentals of Data Science and Analytics

(Common to Computer Science and Business Systems) · Regulations 2021

Time : Three hours

Maximum : 100 marks

Answer ALL questions · PART A — (10 × 2 = 20 marks) · PART B — (5 × 13 = 65 marks) · PART C — (1 × 15 = 15 marks)

Topic repetition frequency — 3 papers analysis

Confusion Matrix / Classification		3/3 ✓
Central Limit Theorem		3/3 ✓
t-test vs ANOVA comparison		3/3 ✓
Survival Analysis definition		3/3 ✓
Facets of Data / Data Science Process		3/3 ✓
Regression (Linear / Logistic / Multiple)		3/3 ✓
Normal Distribution numerical problems		3/3 ✓
EDA techniques		3/3 ✓
Chi-square test (college desires data)		2/3
t-test numerical (library book problem)		2/3
Normal distribution Part C (cold sufferers)		2/3
Time Series Analysis		2/3

PART A — (10 × 2 = 20 Marks)

Unit I — Data Science Fundamentals & EDA

1. What is brushing and linking in exploratory data analysis?
2. Differentiate between structured and unstructured data with examples.

Unit II — Distributions & Statistics

3. How does regression toward the mean differ from other parameters? Give an example.
4. What are outliers in the data? Outline approaches to detect and deal with outliers.

Unit III — Hypothesis Testing

5. State the Central Limit Theorem.
6. Compare between one-tailed and two-tailed tests. Brief about Type I and Type II errors.

Unit IV — Statistical Tests

7. What is the significance of p-value in hypothesis testing? What does it mean when p-values are high and low?
8. Compare between t-test and ANOVA. Define one-factor ANOVA.

9. Why do we need Goodness of Fit? What happens when goodness of fit is low?
10. What is survival analysis? Why do we need weighted resampling?

PART B — (5 × 13 = 65 Marks)

II. (a)

Very High — 93/100

Unit I — Data Science Fundamentals (3/3 papers — always QII)

- (i) Explain in detail about the benefits and uses of data science with counter examples. (6)
- (ii) Describe in depth about Exploratory Data Analysis (EDA) techniques — brushing, linking, smoothing, and rescaling with examples. (7)

Or

(b)

Very High — 92/100

- (i) Illustrate in detail about different Facets of data with examples (Quantitative, Categorical, Ordinal, Date-time, Geographical). (7)
- (ii) Draw and outline the step-by-step activities in the Data Science Process. (6)

12. (a)

Very High — 95/100

Unit II — Descriptive Statistics (appears in ALL 3 papers as Q12)

- (i) The frequency distribution for the length (in seconds) of 100 telephone calls is given below. Compute mean, median, and variance. (7)

Time (sec)	Frequency
0-20	0
21-40	5
41-60	7
61-80	14
81-100	28
101-120	21
121-140	13
141-160	9
161-180	3

- (ii) Consider 15 student exam scores: 2, 17, 5, 3, 28, 7, 5, 8, 5, 6, 2, 12, 10, 4, 3. Find mode, median, mean and identify skewness. (6)

Or

(b)

Very High — 90/100

- (i) SAT math scores: mean=500, SD=100. Find probabilities for: (1) More than 570 (2) Less than 515 (3) Between 520 and 540. (6)
- (ii) Wind speed X (mph) and wave height Y (feet): Create scatter plot and find least squares regression equation. Predict Y when X=15. (7)

X (Wind)	0	2	7	9	13	22
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13. (a)

Very High — 91/100

Unit III — Confidence Interval & Sampling Distribution

- (i) Reading achievement scores for 64 fourth graders, population $SD=0.4$, sample mean= 3.82 . Construct 95% confidence interval and interpret. (6)
- (ii) Illustrate in detail about estimation methods and confidence interval — point estimate, interval estimate, margin of error, Z-interval, t-interval. (7)

Or

(b)

Very High — 89/100

- (i) Exemplify in detail about the significance of z-test, its procedure and decision rule with a numerical example. (6)
- (ii) Simple population: 2, 4, 6, 8, 10. List all samples of size two. Construct sampling distribution of the mean. (7)

14. (a)

Very High — 92/100

Unit IV — t-test + Chi-square (EXACT same problems in 2/3 papers)

- (i) A library lends books for 21 days. Sample of 8 records: 21, 15, 12, 24, 20, 21, 13, 16. Test $H_0: \mu=21$ using t-test at 0.05 level. (7)
- (ii) 90 college students: desires are Love(25), Wealth(10), Power(5), Health(25), Fame(10), Family Happiness(15). Test if desires are equally popular using chi-square test at 0.05. (6)

Or

(b)

Very High — 90/100

- (i) Illustrate one-factor ANOVA with complete numerical example. Show ANOVA table with SS, df, MS, and F values. (7)
- (ii) Brief about Tukey's HSD test and explain two-factor ANOVA. (6)

15. (a)

Very High — 91/100

Unit V — Logistic Regression + Linear Regression numerical

- (i) Describe logistic regression model in predictive analysis — sigmoid function, log-odds, decision boundary, comparison with linear regression. (7)
- (ii) Electronic device lifetime vs temperature. Find linear regression equation. Estimate lifetime at 55°C . (6)

Temp ($^\circ\text{C}$)	10	20	30	40	50	60	70	80	90
Lifetime (hrs)	420	365	285	220	176	117	69	34	5

Or

(b)

Very High — 90/100

- (i) Illustrate time series analysis — components (Trend, Seasonal, Cyclic, Irregular), moving averages, and forecasting methods with examples. (7)

PART C — (1 × 15 = 15 Marks)

16. (a)

Very High — 94/100

Normal distribution multi-part problem — EXACT same question in 2/3 papers. $\mu=83$, $\sigma=20$.

An investigator polls common cold sufferers estimating hours of physical discomfort. Assume normal distribution with mean = 83 hours and standard deviation = 20 hours.

- (i) What is the estimated number of hours for the shortest-suffering 5 percent? (3)
- (ii) What proportion of sufferers estimate their colds lasted longer than 48 hours? (2)
- (iii) What proportion suffered for fewer than 61 hours? (2)
- (iv) What is the estimated number of hours suffered by the extreme 1 percent either above or below the mean? (2)
- (v) What proportion suffered for between 1 and 3 days (24 to 72 hours)? (3)
- (vi) What proportion suffered for between 2 and 4 days? (3)

Or

(b)

High — 82/100

University admission GPA follows normal distribution with mean = 3.20, SD = 0.30.

- (i) $\text{GPA} \geq 3.50 \rightarrow$ automatic admission. What proportion qualifies? (4)
- (ii) $\text{GPA} \leq 2.50 \rightarrow$ automatic denial. What proportion is denied? (3)
- (iii) Honors program: $\text{GPA} \geq 3.75$. What proportion is eligible? (4)
- (iv) If limited to top 10%, what minimum GPA is required? (4)

Pattern summary — key exam insights

- ✓ Q11 — Data Science fundamentals + EDA $\rightarrow$ 3/3 papers always here
- ✓ Q12 — Mean/median/variance from frequency table $\rightarrow$ 2/3 papers same data
- ✓ Q13 — Confidence interval + z-test $\rightarrow$ every year
- ✓ Q14 — Library t-test + college desires chi-square $\rightarrow$ exact same data 2/3 papers
- ✓ Q15 — Logistic regression + Time series (rotating)
- ✓ Q16 — Cold sufferers normal distribution $\rightarrow$ EXACT same in 2 papers. Very high chance!